

Bugs found: 5

1. Potential deadlock: two threads acquire locks A and B in opposite order, which can cause a circular wait.
2. Integer overflow: adding `Integer.MAX_VALUE` to itself overflows silently, producing a negative result.
3. `ClassCastException`: casting a `String` object to `int` via `(int)` is illegal and will throw a runtime exception.
4. Unsynchronized access to shared mutable list 'cache': it is written in main and may be read by threads without synchronization.
5. Thread safety issue: the `Thread` objects are started but never joined or managed, risking resource leaks and unpredictable shutdown behavior.